GG-W-00253A (DSA-DM)
January 22, 1973

SUPERSEDING
Revision of
Int. Fed. Spec. GG-W-00253(GSA-FSS)
April 9, 1962

INTERIM FEDERAL SPECIFICATION

WHEELS, ABRASIVE, IMPREGNATED RUBBER, PUMICE

This interim Federal specification was developed by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers abrasive wheels suitable for use in dentistry for polishing metals, plastic and porcelain.

1.2 Classification.

1.2.1 Classes and sizes. Abrasive wheels shall be of the following classes and sizes, as specified (see 6.1):

Class 1.	Knife edge.
Size 1.	3/8 inch diameter
Size 2.	5/8 inch diameter
Class 2.	Round or square edge.
Size 1.	7/8 inch diameter

FSC 6520

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2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Boxes, Setup.

STANDARDS

Federal

FED-STD-123	Marking for Domestic Shipment (Civilian Agencies).
FED-STD-601	Rubber, Sampling and Testing

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.)

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle and Washington, D. C.)

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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SPECIFICATIONS

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

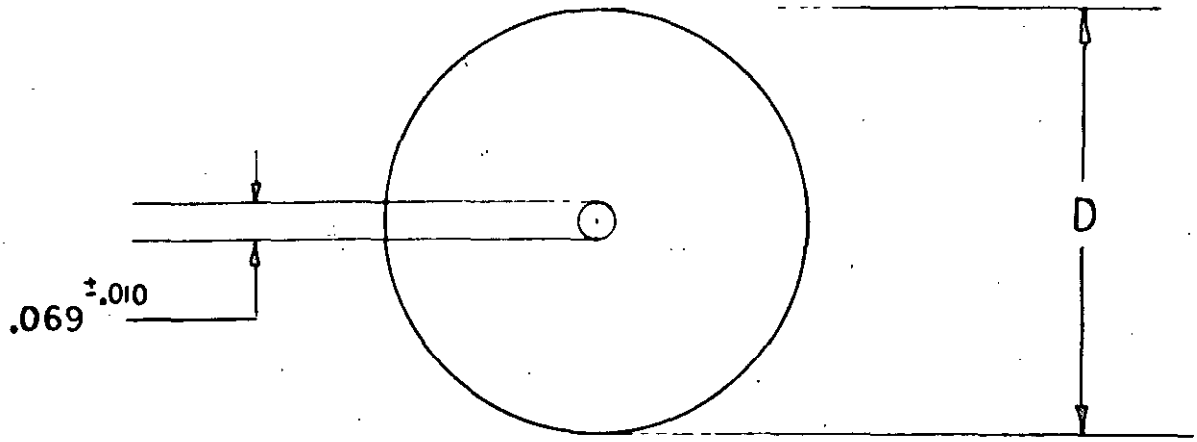
3.1 Material. The wheels shall be made from a suitable rubber compound into which has been incorporated 70 ± 5 percent of grade 1½ ground pumice, as ascertained from the manufacturer's batch formula control records. The pumice shall be free from quartz, obsidian and iron, and shall be odorless and tasteless. A 100-mesh ground sponge may be incorporated into the compound provided the wheels meet the requirements specified hereinafter.

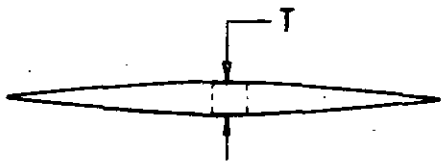
3.2 Style, design and dimensions. The wheels shall be of the style, design, and dimensions shown in Figure 1 for the applicable class and size.

3.3 Working qualities. The wheels shall possess such flexibility that they will conform to the applied surfaces when revolving under high or low speed, and shall be suitable for polishing metals, plastic and porcelain, when tested as specified in 4.4.1.

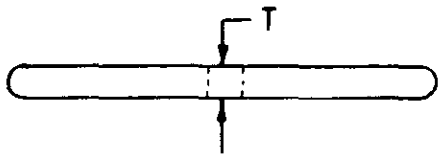
3.4 Accelerated aging. The wheels shall remain flexible and suitable for use in dentistry for polishing metals, plastic and porcelain, after being subjected to the accelerated aging test specified in 4.4.2.

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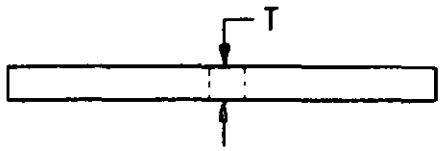




KNIFE EDGE
(Class 1, Size 1 or 2)



ROUND EDGE
(Class 2, Size 1)



SQUARE EDGE
(Class 2, Size 1)

Class	Size	Diameter of Wheel "D"	Thickness of Wheel At Center "T"
1	1	0.375 ± 0.012	0.095 ± 0.010
1	2	0.625 ± 0.012	0.119 ± 0.010
2	1	0.875 ± 0.012	0.125 ± 0.010

Note: Dimensions are in inches

FIGURE 1

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3.4.1 Life expectancy. As an alternate to paragraph 3.4, the wheels shall remain flexible and suitable for use intended, for not less than 3 years from date of delivery to the Government, when stored at controlled room temperature (59° to 86° Fahrenheit). Supplier shall submit certification with regard to life expectancy. When inspection is made at source, the certificates shall be submitted to the cognizant inspector for transmittal to the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, Attention: DPSC-ATQ. When inspection is made at destination, certificates shall be submitted by the contractor directly to the Defense Personnel Support Center. (Note: Submittal of certificates to the Defense Personnel Support Center is applicable only to Defense Personnel Support Center procurements).

3.5 Scratch. The wheels shall cause no pronounced scratches, when tested as specified in 4.4.3.

3.6 Delivery schedule. Not more than 3 months shall have elapsed from date of manufacture of the abrasive wheels to date of delivery to the Government.

3.7 Workmanship. The wheels shall have a tumbled finish and mandrel holes shall be centered in wheels. Wheels shall be free from defects which detract from their appearance or which may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

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4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the materials may be furnished in lieu of actual performance of such testing by the contractor, provided that lot identity has been maintained and can be demonstrated to the Government. The Certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the item identification, the name of the material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Quality conformance inspection. The examinations and tests required to assure conformance of those items or lots of items to be offered are classified as specified in tables I and III (see 4.2.1 and 4.3) and table II (see 4.2.2.).

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels (AQL's) shall be as indicated in table I. Unit of product for examination purposes shall be one unit of issue.

Table I Sampling for Examination

Type Defect	Inspection Level	AQL (Percent Defective)
Classification of defects		
Major	II	1.5
Minor	II	4.0
Dimensional examination	S-2	4.0

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and Table II, except that the acceptance number shall be zero for all sample sizes. The unit of product for test purposes shall be one unit of issue.

Table II Test criteria

Characteristic	Requirement	Test procedure	Inspection level
Working qualities	3.3	4.4.1	S-1
Accelerated aging	3.4	4.4.2	S-2
Scratch	3.5	4.4.3	S-2

4.3 Examination. The wheels shall be examined to determine compliance with the requirements of this specification. Examination shall be conducted in accordance with the following classification of defects. Examination shall include, but shall not be limited to, the defects listed in table III.

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Table III Classification of defects

Categories	Defects
Major	
101	Wheels not flat.
102	Mandrel holes not centered in wheels.
103	Wheels do not have tumbled finish.
104	Wheels do not have knife edge, or do not have round or square edge, as specified.
Minor	
201	Packaging and packing not as specified.
202	Marking on packaging or packs incorrect, illegible, incomplete, or missing altogether.
203	Pumice not odorless and tasteless.

4.4 Test procedures.

4.4.1 Working qualities. Wheels shall be subjected to a practical operating test to determine conformance with 3.3.

4.4.2 Accelerated aging test. Abrasive wheels shall be subjected to the aging test in accordance with method 7221 of Federal Test Method Standard 601.

4.4.3 Scratch test. Place an abrasive wheel on a clean plate glass. Rub all the critical working surfaces of the wheel over the surface of the glass for 3 minutes, using circular and straight strokes. Examine surface of the glass for any pronounced scratches.

5. PREPARATION FOR DELIVERY.

5.1 Packaging. Packaging shall be Level A or C as specified (see 6.1).

5.1.1 Unit of issue. One box, containing 25 or 100 wheels, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table IV shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

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Table IV Packaging Quantities

Class	Size	Unit package	Intermediate package	Exterior container
1	1	1 unit	12 units	1728 units
1	2	1 unit	12 units	576 units
2	1	1 unit	10 units	120 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Unit package shall be a box of appropriate size and design constructed in accordance with PPP-B-566 or PPP-B-676, except commercial colors will be acceptable. Closure shall be adequate to prevent accidental opening under normal handling.

5.1.3.2 Intermediate package. Intermediate package shall be a box of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676, except that commercial colors will be acceptable, or PPP-B-636. Closure shall be adequate to prevent accidental opening.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Levels A, B or C as specified (see 6.1).

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification. Special requirements of applicable table shall apply for PPP-B-636 box.

5.2.1.2 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

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5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Fiberboard box shall be closed by method II of PPP-B-636. Special requirements of applicable table shall apply for PPP-B-636 box.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Civil agencies. Marking for shipment shall be in accordance with FED-STD-123.

5.3.2 Military activities.

5.3.2.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Class and size required (see 1.2.1).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.2 Specification changes. Because of the extensiveness of changes to this document, the margins have not been marked with an asterisk to indicate where changes from the previous issue have been made.

6.3 Limitation. This specification does not cover all types, classes, numbers, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover those items which are normally procured to meet Federal Government requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6520-1346

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE: GG-W-00253A(DSA-DM)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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